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An Innovative Approach to Predict Viable Pore Pressure in Carbonate Rocks in the Middle East

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Abstract

Abnormal pore pressures have been frequently observed in various regions globally, particularly in exploration areas. Accurate Pore Pressure Prediction and designing safe mud weights during drilling execution are crucial for ensuring safety and efficient operations. This paper presents an innovative approach to predict pore pressure prediction in carbonate reservoir, analyzing the elastic properties changes after prolong production/injection and incorporate an adjustment factor to predict the pore pressure that matches with actual measurement

The primary goal of pore pressure prediction is to provide pressure data before drilling operations. Historical studies indicate that pore pressure prediction equations based on sonic, resistivity, and porosity are not effective in carbonates (O' Connor et al, 2010). However, the current study confirms that the pore pressure correlates greatly with interval velocities, therefore pore pressure prediction in carbonate rocks can be quite accurate if the parameters are optimized and adjustment factor is applied. This paper presents a workflow that discusses data gathering & quality control for fifty wells, logs correction, sensitivity scenarios, overburden and hydrostatic gradient estimation to predict pore pressure within carbonate reservoirs. The paper also covers modelling part to predict the expected change in elastic properties after prolong production/injection to calculate the adjustment factor for accurately estimating pore pressure that matches with actual measurements.

Choosing the appropriate parameters in Eaton's equation resulted in a fairly accurate transformation from velocity to pore pressure. The sonic logs for fifty wells have been corrected for drift and invasion. Based on the actual samples taken from three formations, salinity appears to change drastically from shallow to deeper formations and therefore the hydrostatic pressure gradients. For pore pressure prediction, an average value of 0.485psi/ft has been used. The predicted pore pressure in the overburden shows 98% match with measured pressures without incorporating adjustment factor. This is because the pore pressure and the logs were acquired on the same date. In the reservoir section, the predicted pore pressure shows 150psi to 200psi difference with initial and last measured pressures respectively. Therefore, an adjustment factor is needed which depends on the cumulative injection/production in vicinity and it is estimated through fluid substitution modelling. The modelling results shows that V_p (Primary velocity) increases by 6.5% when seawater displaces 90% oil in the reservoir section. Based on this, an adjustment factor has been designed

and incorporated in pore pressure prediction equation to estimate the current reservoir pressure that can be used for Mud Weight (MW) designing. The resultant pore pressure values show fair to good correlation with the history matched model and actual measurements from the well.

To address pore pressure challenges in the Middle East carbonate reservoirs, the optimized parameters have been chosen along with an adjustment factor that work effectively, thereby minimizing drilling risks in overburden and aiding in drilling long lateral horizontal wells within the reservoir section.

Introduction

Pore pressure prediction is a critical aspect of drilling operations, especially in carbonate reservoirs where abnormal pore pressures are often encountered. The drilling records showed that drilling complexities such as stuck pipes, hole collapse and kicks, were encountered during drilling, and these events were highly related to abnormal pore pressure. Accurate prediction of pore pressure helps in designing safe mud weights, preventing blowouts, and ensuring the overall safety and efficiency of drilling operations. The high pore pressure results in a very narrow mud weight window and as a result with high drilling risks. If the drilling mud weight is lower than the formation pressure, it can lead to borehole collapse, kicks, and blowout et al (Gao et al, 2025).

Historical studies suggest that some pore pressure prediction methods have been developed using geological data, including seismic, well logs, and drilling data (Eaton, 1975; Bowers, 1995; Dutta, 2002). Some other techniques include pore pressure prediction with sonic while drilling (Cai et al, 2015a and 2015b; Gao et al, 2019), pore pressure prediction with seismic while drilling (Guo et al, 2018 and Gao et al, 2020), pore pressure prediction with bed boundary detection (Guo et al, 2020) and pore pressure prediction with borehole vertical seismic profile (Wang et al, 2020 and Gao et al, 2021). This paper aims to provide a comprehensive approach to pore pressure prediction using Eaton's method, focusing on the parameter's optimization in carbonate reservoirs, the analysis of elastic properties changes after prolonged production/injection and incorporating an adjustment factor to predict accurate pore pressure that matches with actual measurements.

Reservoir pore pressure is dynamic which will vary with production and injection in the reservoir. The assessment of the pressure within reservoir is critical for drilling long horizontal wells. Simulation models are always a quick way to understand pressure however they are based on injection/production and then calibrated with actual pressure measurements. Uncertainty remains in the model and that's why most of the wells are planned to be drilled with pressure measurement tools along the drainage section. The pressure analysis at well location based on logs usually doesn't work well in carbonates. One of the main reasons is that pressure data is dynamic, and logs are static. To match both, an adjustment factor of injection/production need to be incorporated. The current study minimizes this difference in dynamic verses static properties by designing an adjustment factor, derived from fluid substitution modeling.

The study confirms that pore pressure correlates significantly with interval velocities, enabling accurate predictions in carbonate rocks when parameters are optimized, and an adjustment factor is applied. The paper outlines a workflow involving data gathering and quality control for fifty wells, log corrections, sensitivity scenarios, and estimations of overburden and hydrostatic gradients. It also includes modeling to predict changes in elastic properties after prolonged production or injection, which helps calculate the adjustment factor for accurate pore pressure estimation. The predicted pore pressure in the overburden shows 98% match with measured pressures without incorporating adjustment factor. This is because the pore pressure and the logs were acquired on the same date. In the reservoir section, the predicted pore pressure shows 150psi to 200psi difference with initial and last measured pressures respectively. Therefore, an adjustment factor is needed which depends on the cumulative injection/production in vicinity and it is estimated through fluid substitution modelling. Fluid substitution modeling indicated a 6.5% increase in Vp (primary velocity) when seawater displaced 90% oil in the reservoir section. The adjustment factor was then

designed and incorporated into the pore pressure prediction equation to estimate current reservoir pressure for Mud Weight (MW) design, minimizing the difference to 10psi and showing good correlation with the history-matched model and actual measurement.

Geological Settings

The study area is located offshore in the Middle East. Structurally, most of the oil producing fields are low dip (~1-2 degrees) anticlines. The overburden section comprises a mix of anhydrite, calcite and dolomite. Rocks in the overburden are mainly saline aquifers. The reservoir rocks consist mainly of limestones, separated by dolomite layers. Some of the reservoirs are fractured. Present day subsurface structures are formed mainly due to two tectonic episodes (Fig.1); one in Late Cretaceous and second in Late Oligocene (Ahmad et al, 2022). Main faults occurred in the Late Cretaceous age. Primarily, faults trend in NW-SE direction and extend several kilometers. The tectonic stresses turned orthogonal (NE-SW) in Late Oligocene age which further deformed subsurface structure. As a result, subsurface complexity increased many folds due to stress orientation changes over time.

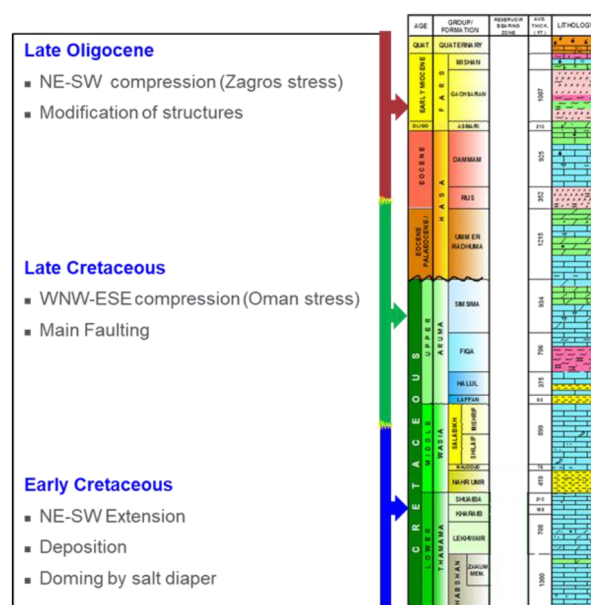


Figure 1—Generalized lithostratigraphic column. Text on the left label's important tectonic episodes ([Ahmad et al, 2022](#))

Velocity Based Pore Pressure Prediction

It is common practice in siliciclastic velocity-based pore pressure prediction to relate shale properties with effective stress and hence the relationship between rock properties and stress is simpler to predict for shales (Wodu et al, 2014). On the other hand, the historical studies also highlight that the pore pressure prediction equations based on sonic, resistivity and porosity are available in the literature but do not work well in carbonates (O' Connor et al, 2010).

The current study focuses on velocity-based pore pressure prediction in carbonate rocks using Eaton's equation. The prediction of pore pressure is divided into two sections: overburden and reservoir. In the overburden section, only one well has been analyzed, with both sonic and actual pressure measurements taken simultaneously. The estimated pore pressure based on Eaton's equation shows a 98% match with actual measurements, which is expected since both pressure points and logs were acquired on the same date, with no change in subsurface fluid effects. However, in the reservoir section, 50 vertical and slightly deviated wells, mostly drilled 25-30 years ago, were analyzed. Sonic logs were used to calculate interval velocities, evaluate data to estimate appropriate hydrostatic and overburden pressures, and predict pore pressure using

Eaton's equation. An adjustment factor was designed based on expected fluid replacement within the reservoir and incorporated into the pore pressure prediction equation to estimate current reservoir pressure. Pore pressure within the reservoir has been estimated for two main oil-bearing reservoirs (reservoir-2 and reservoir-3). Reservoir-2 has more well penetration, leading to more reliable results compared to reservoir-3.

The overall methodology for both overburden and reservoirs remain the same and it involves several key steps:

1. Data gathering & quality control
2. Log's corrections
3. Hydrostatic pressure estimation based on the actual fluid sample
4. Overburden pressure estimation from density logs
5. Interval velocity estimation in each reservoir section, converting them into points and mapping
6. Pore pressure prediction using Eaton's equation for overburden and reservoir
7. Pore pressure mapping and validation
8. Adjustment factor incorporation based on injection or production in vicinity

Data Gathering & Quality Control

Initially, data from approximately fifty wells were collected, including formation tops, composite well logs, survey data, wireline logs, fluid samples and direct pressure measurements. The geological setting of the field and various issues encountered during drilling, such as dynamic mud losses or flow, were considered during the evaluation.

Significant time was dedicated to data quality control (QC) for all fifty wells, which involved correcting logs for invasion and despiking. High-frequency sonic velocities were compared with Vertical Seismic Profile (VSP) velocities, and drift was estimated and corrected (Fig.2). Well logs were then correlated and well tops were refined to ensure consistency within the reservoirs of interest. Additionally, density logs were corrected based on borehole conditions to ensure they accurately represent subsurface conditions before being used as inputs for pore pressure prediction.

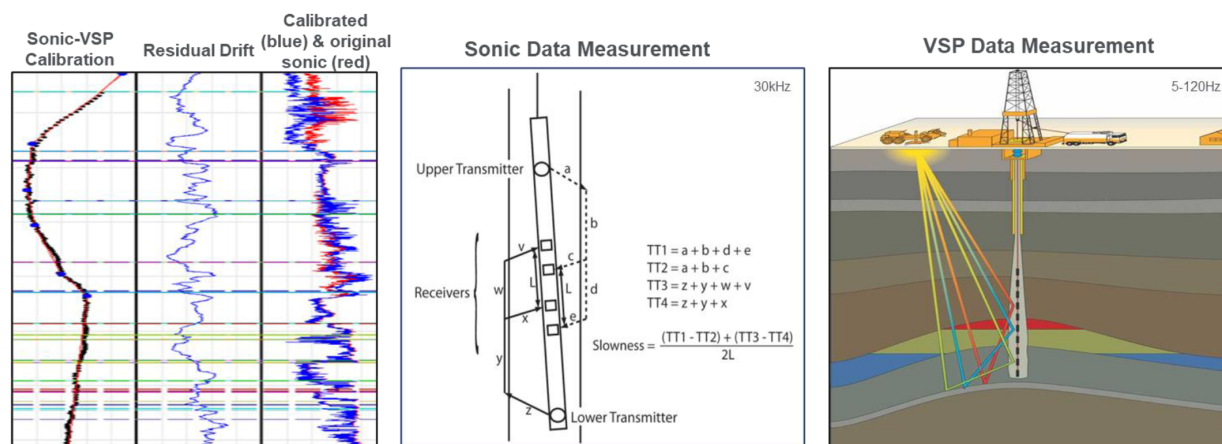


Figure 2—The figure displays VSP and Sonic data measurements and compares their travel time. The plot on the left shows the calibration of travel times for both methods by minimizing residual drift and estimating calibrated sonic logs.

Hydrostatic Pressure Estimation

Hydrostatic Pressure (HP) is the pressure exerted by the weight of a static column of pore fluid connected to the atmosphere (Wodu et al, 2014). While the average seawater gradient is 0.443 psi/ft, samples from different formations show that water salinity can reach up to 300,000 ppm (Fig.3), causing variations in

the gradient value. For the current analysis, an average gradient of 0.485 psi/ft has been used, based on gradients established from measured pressure data and sample analysis.

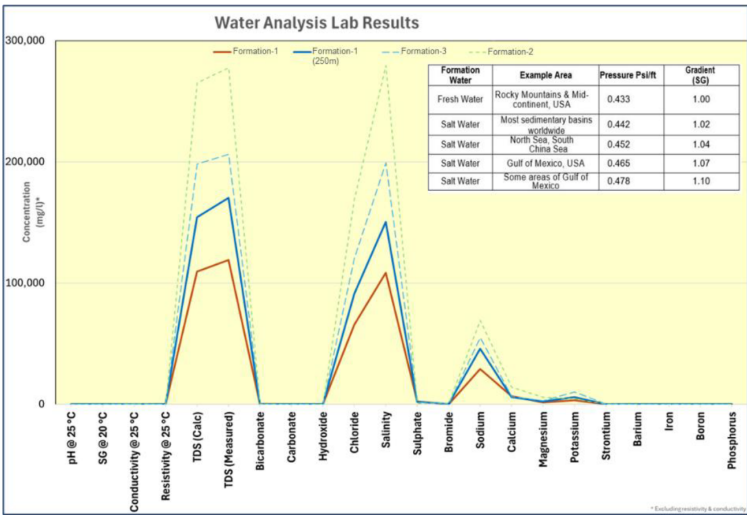


Figure 3—Water sample analysis, showing varying salinity of different formation fluids. Salinity appears to vary both laterally and vertically.

Overburden Pressure Estimation

The Overburden Pressure (OBP), vertical or lithostatic stress at any given depth is the pressure exerted by weight of overlying column of seawater (if any), and sediments including the weight of the contained fluids (Wodu et al, 2014). OBP was calculated by integrating the best-fit density trend derived from the edited and despiked density log data. The details of empirical equations describing bulk rock density as a function of depth below the mud line are provided by Wodu et al, 2014. Based on the analysis, an overburden gradient of 1.01psi/ft is deemed optimal for use in pore pressure prediction (Fig.4).

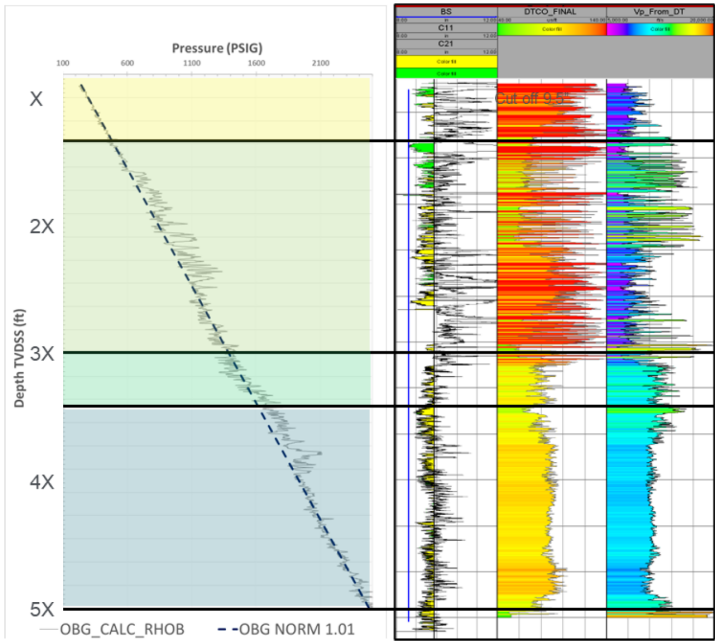


Figure 4—Overburden pressure estimation from the density log. An overburden gradient of 1.01psi/ft is deemed optimal for use in pore pressure prediction

Results and Discussion

Overburden Section

For the estimation of pore pressure in the overburden section, the following parameters have been used:

Table 1—Parameters used for pore pressure prediction

Eaton's sonic exponent	1.1
Hydrostatic pressure gradient	0.485
OBG	1.01

The hydrostatic pressure gradient value of 0.485 psi/ft is the average number used for all overburden formations.

The resultant pore pressure curve (grey) in Fig. 5 shows a good overall match with actual measured pressure points (black). The mud gradient line (green) indicates a 9.5 ppg MW used to drill the surface hole section, showing that the MW used is slightly overbalanced. In formation-1 and formation-2, both estimated and measured pressure points are aligned, though there are some fluctuations in formation-3 and formation-4. During drilling, dynamic mud losses were observed in formation-3, which aligns with the estimated pore pressure response. Additionally, this is an expected zone of crossflow between formation-3 and the top zone of formation-4, as evidenced by the salinity drop in formation-3 (Fig.3).

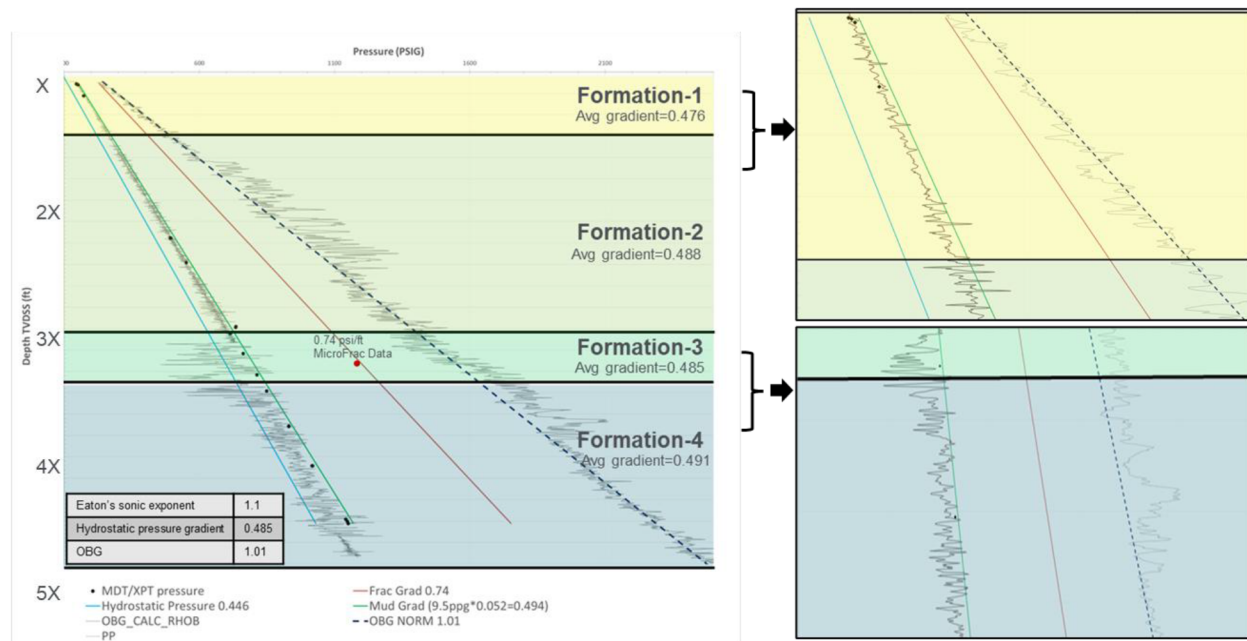


Figure 5—Overburden pore pressure estimation. The estimated pore pressure results match the actual measured pressure because the input logs used for prediction were acquired on the same date as the measured pressure

Historical overburden Frac Gradient (FG) has been estimated based on several offset wells, ranging from 0.7 psi/ft to 0.75 psi/ft. A recent Minifrac measurement in formation-3 shows FG value of 0.74 psi/ft, which aligns with the regional trend (Fig. 5).

The estimated pore pressure results match the actual measured pressure because the input logs used for prediction were acquired on the same date as the measured pressure. Consequently, no further modeling was performed.

Reservoir Section

In the reservoir section, the average interval velocities for all fifty wells within Reservoir-2 were calculated, and a velocity map was prepared to check for variations and identify any anomalies. No anomalies were observed, confirming that the log conditioning was done effectively. The predicted pore pressure from interval velocities using Eaton's equation shows a strong correlation with interval velocities in Reservoir-2 and Reservoir-3, although there is a slight change in trend, as shown in the cross plot in Fig. 6. This interval velocity map was then used as an input to estimate the pore pressure map for Reservoir-2 (Fig.6) using the regression line method from the pore pressure and interval velocity cross plot.

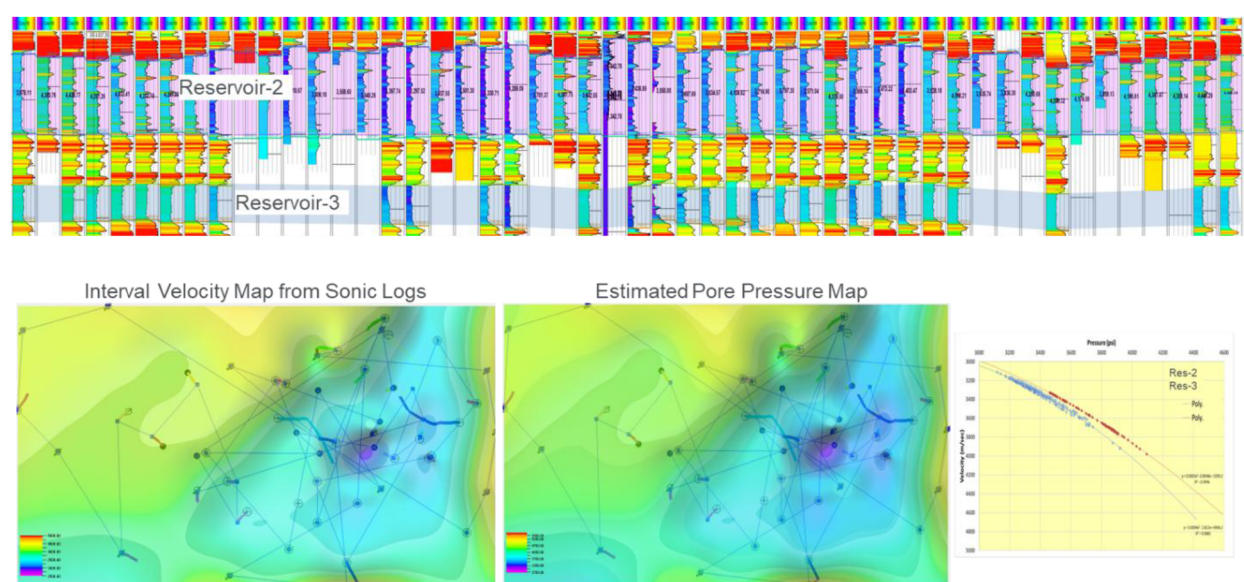


Figure 6—Reservoir pore pressure prediction (Reservoir-2). The average interval velocities were calculated for fifty wells, and a velocity map was prepared which was converted to pore pressure map using regression line from the cross plot.

To understand the effect of injection and production within reservoir-2, the initial measured pore pressure values for all available wells were gathered and plotted against the pore pressures measured in the same wells at later stages (Fig. 7). Ideally, without any interference from injection or production, the initial pressure versus pressure measured at any later stage would form a straight line. However, the cross plot in Fig. 7 shows a scattered trend. Points above the ideal case trend line represent pressure drops due to production from the reservoir, while points below the trend line indicate pressure build-up due to injection. It is observed that injections have a greater effect on reservoir pressure than production. Additionally, most of these scattered points are aligned along the initial reservoir pressure (3850 psi).

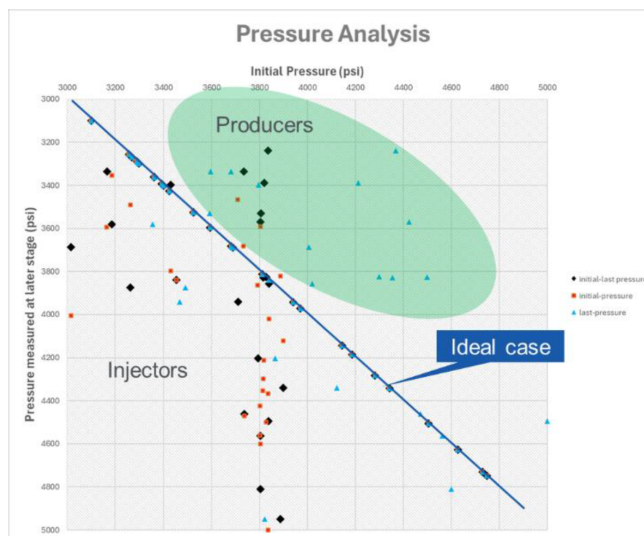


Figure 7—Initial measured pore pressure values for all available wells are plotted against the pore pressures measured in the same wells at later stages which shows a scattered trend. Production and injection trends are labeled.

To demonstrate the effect of individual wells across the field, the initial and last measured pressure points were mapped separately using the Moving Average algorithm with consideration of isotropic reservoir conditions. A difference map was then created to evaluate the change in measured pore pressure at later stage (initial-Last in Fig. 8). The histogram shows that 56% of the points have a difference of up to ± 100 psi, and 20% of the points have a difference of up to 200 psi. Another difference map was created between the pore pressure predicted using Eaton's equation for Reservoir-2 in Fig. 6 and the initial measured pore pressure map (initial minus estimated), showing that only 24% of the points have a difference of up to ± 100 psi. However, 40% of the points show a pressure variation of up to +150 psi, which is expected due to injection (Fig. 8).

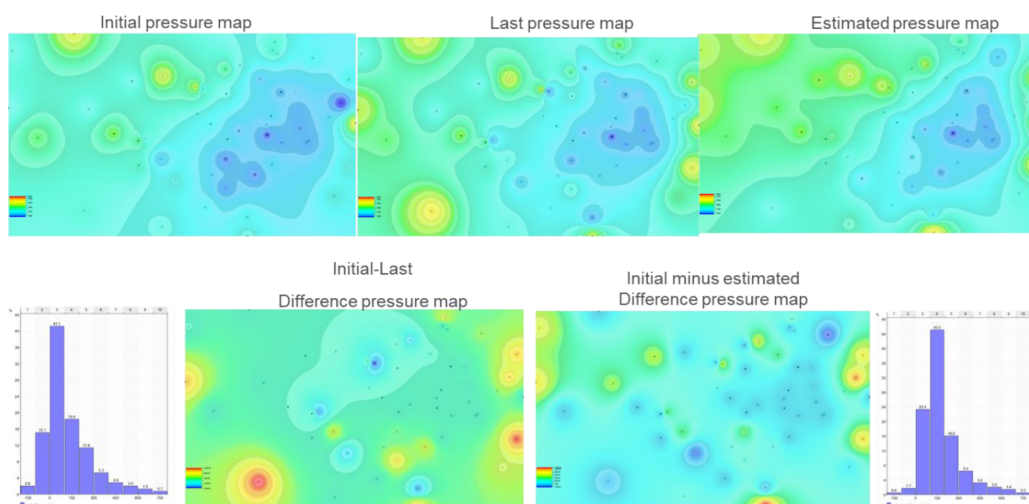


Figure 8—Initial and last measured pressure points were mapped separately and a difference map was then created. Another difference map was created between the initial measured pressure and the estimated pressure, showing 24% of the points with difference of up to ± 100 psi.

The estimated pressures are also compared with the initial history-matched modeled pressure (blue lines) and the current history-matched modeled pressure (green lines) (Fig. 9). The results indicate that the estimated pressure aligns more closely with the initial modeled pressure than with the current modeled

pressure. Therefore, to estimate pore pressure that reflects current dynamic conditions, an adjustment factor must be applied to Eaton's equation. This adjustment is achieved through fluid substitution modeling.

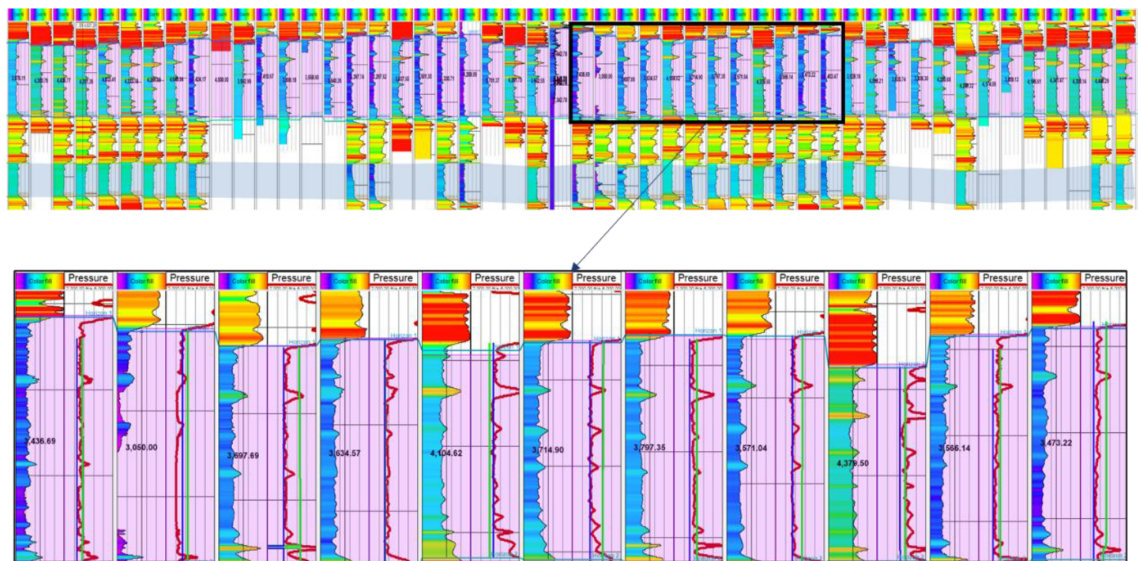


Figure 9—The estimated pressures are compared with the initial history-matched modeled pressure (blue lines) and the current history-matched modeled pressure (green lines).

The expected changes in elastic properties after prolonged production or injection are modeled to calculate the adjustment factor for accurately estimating pore pressure that aligns with actual measurements under current dynamic conditions.

The fluid substitution modeling results show that V_p increases by 6.5% when 90% of the oil is displaced by seawater, while it decreases by 1.5% when oil is displaced by gas in reservoir-2 (Fig. 10). Since interval velocities strongly correlate with pore pressures (Fig. 6), we can infer the percentage of pressure change from the velocity-pressure cross plot. In the current case, it is approximately 5%, although this varies for individual wells based on the total injection or production to date.

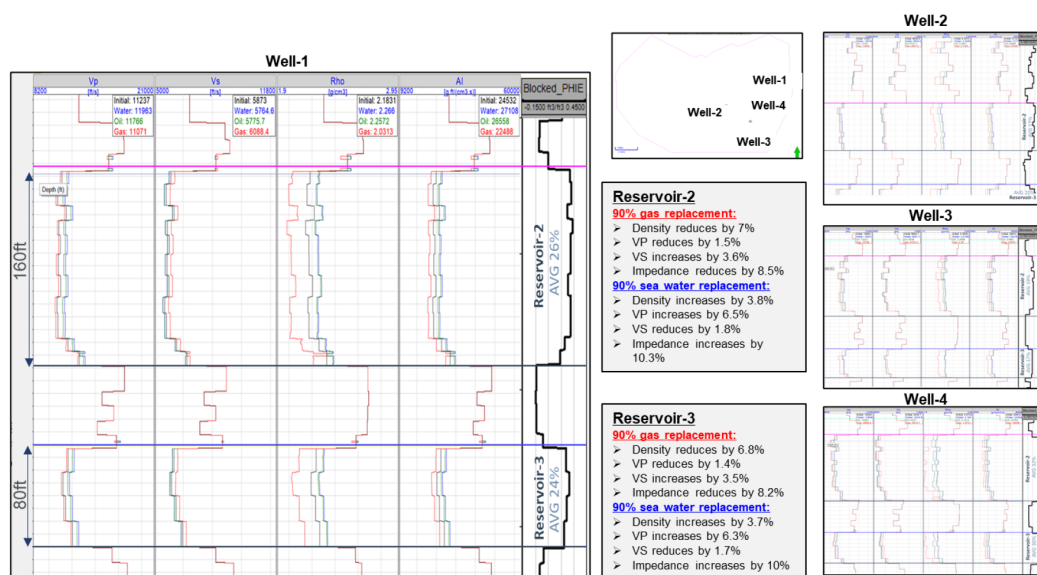


Figure 10—The fluid substitution modeling results show that V_p increases by 6.5% when 90% of the oil is displaced by seawater, while it decreases by 1.5% when oil is displaced by gas in reservoir-2

When this 5% change in pressure is applied as an adjustment factor to the estimated pressure, the estimated pressures now match the last measured pressures within reservoir-2, with a difference of up to 10 psi only.

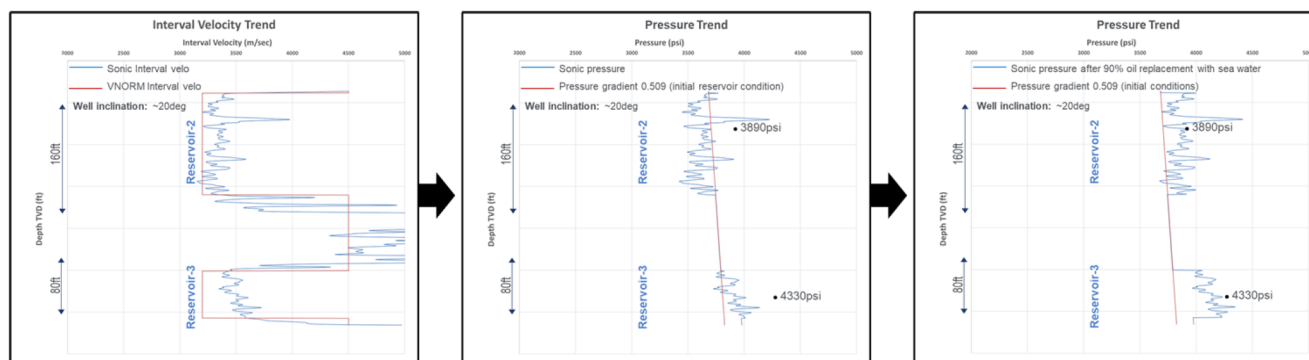


Figure 11—An adjustment factor was estimated based on 5% change in pressure after injection. The corrected pressure now matches with last measured pressure.

Overall, the study demonstrates that selecting the appropriate parameters in Eaton's equation leads to a fairly accurate transformation from velocity to pore pressure, matching the initial pressure. However, by incorporating the adjustment factor, current dynamic conditions can be estimated with greater accuracy.

For future research in pore pressure prediction in carbonate reservoirs, our focus is on utilizing machine learning to enhance the analysis of large datasets and quickly estimate adjustment factors for all wells. Additionally, we are leveraging the existing velocity model to predict pore pressure and gain a better understanding of vertical changes.

Conclusion

The study provides a comprehensive approach to pore pressure prediction in carbonate reservoirs using Eaton's equation. By optimizing the parameters and analyzing the changes in elastic properties after prolonged production/injection, accurate pore pressure prediction can be achieved. This helps in minimizing drilling risks and ensuring the safety and efficiency of drilling operations in carbonate reservoirs.

In a mature oil field, reservoir pore pressure changes with production and injection, making accurate pressure assessment crucial for drilling long horizontal wells. Simulation models help understand pressure quickly but require calibration with real measurements, leaving some uncertainty in actual subsurface properties. Thus, most wells use pressure measurement tools during drilling. Pressure analysis using logs is often ineffective in carbonates because pressure data is dynamic, while logs are static. To reconcile this, an adjustment factor based on fluid substitution modeling is used. The study shows that using the Eaton equation with sonic logs provides reasonable pore pressure predictions in both overburden and reservoir.

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Nomenclature

- HP = Hydrostatic Pressure
- FG = Frac Gradient
- MW = Mud Weight
- OBP = Overburden Pressure
- Vp = Primary Velocity

VSP = Vertical Seismic Profile

References

- Ahmad, J., Khouri, N., Sarra, A., Khulood, A., 2022, Overburden drilling challenges & their possible mitigations using integrated seismic structure characterization in a giant carbonate field in Middle East, Abu Dhabi International Petroleum Conference and Exhibition, 31 October-03 November 2022, Abu Dhabi, UAE, SPE 211657.
- Bowers, G.L., 1995. Pore pressure estimation from velocity data: Accounting for overpressure mechanisms besides undercompaction. *SPE Drilling & Completion*, **10**(02), 89–95.
- Cai, J., Li, W., Liu, P., Chen, M., Zheng, J., & Cheng, Y., 2015b. Research and application of while-drilling pressure-monitoring technology in exploration wells of deepwater area, Qiongdongnan Basin. *Natural Gas Industry*, **35**(10), 99–105.
- Dutta, N.C., 2002. Geopressure prediction using seismic data: Current status and the road ahead. *Geophysics*, **67**(6), 2012–2041.
- Eaton, B.A., 1975. The equation for geopressure prediction from well logs. In SPE Annual Technical Conference and Exhibition. (pp. SPE-5544).
- Gao, Y., Chen, M., Du, C., Wang, S., Sun, D., Liu, P., & Chen, Y., 2019. Integrated real-time pressure monitoring enabled the success of drilling a HTHP offshore well: a casing study in Ledong Area of Yinggehai Basin, South China Sea. In International Petroleum Technology Conference (p. D031S064R003).
- Gao, Y., Liu, P., Du, C., Chen, M., Chen, Y., 2020. The Application of Seismic while Drilling in High Temperature, High Pressure Reservoirs of the Yinggehai Basin. *Petroleum Drilling Techniques*, **48**(4): 63–71.
- Gao, Y., Chen, Y., Chen, M., Liu, P., Zhang, C., Wang, S., ... & Du, C., 2021. Determine HTHP reservoir depth with advanced acoustic impedance inversion in an offshore well at South China Sea. In Proceedings of the 14th SEGJ International Symposium, Tokyo, Japan, 18-21 October 2021 (pp. 23–26). *Society of Exploration Geophysicists and Society of Exploration Geophysicists of Japan*.
- Guo, S., Gao, Y., Chen, M., Liu, P., Chen, Y., Zhang, L., & Huang, Z., 2018. The first application of seismic while drilling technology in HTHP offshore exploration well of South China Sea. In SPE Asia Pacific Oil and Gas Conference and Exhibition (p. D031S029R001).
- Guo, S., Gao, Y., Chen, M., Shim, Y.H., Wang, C., Liu, P., & Ding, S.J., 2020. Advancement of Drilling Hazard Prevention Practices in HPHT Exploration Well-A Case Study in South China Sea. In SPE Annual Technical Conference and Exhibition.
- Wang, S., Liu, P., Guo, S., Gao, Y., Chen, M., & Du, C., 2020. Drilling Geomechanics Successfully Applied in a High-Temperature High-Pressure Offshore Well, South China Sea. In Offshore Technology Conference Asia (p. D012S001R070).
- Wodu, E., Sarker, R., Itua, O., 2014. Modelling Seismic Velocities & Pressure Cells for Pre-Drill Pore Pressure Prediction and Application in Development Wells in Nigerian Deepwater Turbidites. Paper presented at the SPE Nigeria Annual International Conference and Exhibition, Lagos, Nigeria (SPE-172490-MS)